The role of IT leadership in realizing the return on investment of AI coding assistants.

by

Celumusa H Zulu

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Supervisor: Dr. Siyabonga Mhlongo

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Abstract

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Background

Artificial intelligence (AI) is one of the important structural influencers on productivity growth and competitive advantage today (Gillespie et al., 2025). As of 2023, generative AI systems have progressed from experimentation to operationalization within organizational processes (Gillespie et al., 2025). Recent global studies suggest that organizations are investing more in AI with clear expectations of understanding tangible economic gains, cost savings, and performance improvements (Gillespie et al., 2025). Research on information systems and management has advanced from adoption to value understanding, with a focus on distinguishing organizational value from investment in AI technologies as a function of organizational governance and management capability rather than technological capability (Hossain et al., 2025).

In software engineering, generative AI tools represent one of the most prominent manifestations of AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023).Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

Both organizational culture and leadership have traditionally been acknowledged as factors that impact the success of technology implementation. In the digital domain, it is the values and norms that foster a culture of experimentation, learning, and data-driven decision-making, as well as a willingness to adopt technological changes. New studies have found that digital culture reinforces the association between human AI competencies and innovation outcomes, suggesting that cultural readiness moderates the performance effects of AI investments (An et al., 2024). Likewise, the same is true for employee engagement in AI initiatives, ethical governance, and leadership's role in influencing strategic alignment. While the importance of digital culture and leadership is increasingly acknowledged in the AI adoption domain, the current body of research remains dispersed (An et al., 2024). Most studies analyse AI adoption from a macro-organizational perspective or focus on individual productivity effects, failing to consider the integration of digital culture, leadership, and ROI in software engineering (Peng et al., 2023). ROI is increasingly becoming the sole rationale for investing in enterprise AI initiatives.

African organizations are accelerating the adoption of AI technologies to address structural inefficiencies and resource limitations. Emerging market organizations have reported high levels of workplace AI experimentation and positive attitudes toward AI productivity (Gillespie et al., 2025). However, the extant literature also emphasizes the lack of digital skills pipelines, governance maturity, and leadership ability, among other things, as difficulties associated with the adoption of GenAI technologies (Hughes et al., 2026). In this regard, the importance of organizational culture and leadership cannot be overstated. In the absence of strategic direction, trust, and a learning-oriented culture, the adoption of AI technologies is more inclined toward fragmentation and symbolic adoption than toward creating value. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

In the context of software engineering in South Africa, it provides a relevant setting for understanding how organizations embrace and use advanced AI tools. Research indicates that AI is emerging as a key factor influencing organizational change and shaping work practices and culture as organizations increasingly embrace AI tools across their business processes and operations (Murire, 2024). There are opportunities and challenges to address to leverage organizational performance through AI adoption and alignment with organizational culture and structure, as indicated by research reviews on AI tool adoption in organizations (Romeo & Lacko, 2025). Though the adoption of AI technology affects many aspects of the organization's performance positively, like productivity and output, the effect of AI technology on aspects like the organization's culture and employees' morale and engagement is complicated and depends on many factors, as indicated by the study on the impact of AI tools in workplace settings (Kassa & Worku, 2025). Studies on the effect of generative AI tools on software engineering have indicated that AI technology positively affects organizations by enhancing productivity in coding and other software engineering activities through employees' adoption of AI tools (Alenezi & Akour, 2025).

Leadership and organizational culture play an important role in the interpretation, adoption, and usage of AI technologies by work teams. Research on AI adoption has found that AI-related technology adoption is also dependent on leadership styles that facilitate embedding technology initiatives into organizational culture (Romeo & Lacko, 2025). Research on the adoption of AI-related technology has also found that the adoption of AI-related technology is dependent on the adoption of the organizational culture that is conducive to the adoption of the technology (Kassa & Worku, 2025). Organizational culture research further indicates that cultural factors, specifically norms related to innovation, collaboration, and adaptability, modulate how employees engage with AI tools and integrate them into their work (Murire, 2024). In software engineering teams, where AI coding assistants alter routines and task assignments, leadership and culture therefore likely determine the extent to which AI investments yield returns in productivity, quality, and financial outcomes, making them essential variables in understanding realized returns on investment.

Problem Statement

Generative AI has shifted its focus from experiment deployment to integrating it into the core knowledge work process. Substantial experimental results show that generative AI can significantly improve the productivity and quality of cognitive work outcomes, especially among lower-performing members, thereby affecting the performance distribution within the organization (Noy & Zhang, 2023). Field experiment results show that the application of generative AI in the real world can improve the organization's productivity and influence relevant outcomes, such as task completion and retention, indicating that the ROI on generative AI comes from its application within the organization, not the technology itself (Brynjolfsson et al., 2025). In the software development field, three large-scale field experiments have shown that providing developers with GitHub Copilot can improve task completion and development activities, providing empirical support for the ROI of generative AI in coding assistance (Cui et al., 2026). Controlled experiment results show that developers using AI coding assistance complete programming tasks more quickly and at a higher rate, providing a business case for AI coding assistance tools based on productivity (Peng et al., 2023).

Despite these measurable improvements, existing research evidence consistently shows that the performance impact differs across contexts. Research in software engineering has shown that AI-assisted code generation produced by GitHub Copilot improves speed but also creates variability in correctness, maintainability, and review effort; this shows the value to depend on the management and integration of such tools and systems (Moradi Dakhel et al., 2023). Research on AI governance in organizations has argued that to sustain value, organizations must have structured processes and systems to ensure clear accountability and supervisory leadership to transform experimentation into sustained performance improvement (Mäntymäki et al., 2022). Research on organizational innovation with generative AI has also shown that organizations can obtain different performance impacts depending on whether AI is used for automation or augmentation; this shows the role of strategic leadership and organizational culture in determining the benefits obtained (Holmström & Carroll, 2025). Overall, this existing research evidence shows the role of organizational culture and strategic leadership in determining whether the improvements in productivity due to AI coding assistants can be sustained and converted to measurable return on investment.

The relevance of these mechanisms can be further accentuated in the context of Africa since organizations face structural limitations in skills development, governance, and digital sophistication. An empirical study of AI readiness in Africa identified the level of preparedness and the institutional ability to adopt AI, thus emphasizing the role of organizations in the transition from AI pilot projects to performance results (Isagah & Ben Dhaou, 2024). In the context of South Africa, empirical studies of the financial sector have identified the role of organizational factors such as leadership support and organizational readiness in the adoption of new and emerging technologies, thus emphasizing the relevance of these factors to the achievement of ROI in AI coding assistants in software development teams (Matsepe & Van der Lingen, 2022). Empirical research carried out within the context of South Africa suggests the use of sociotechnical systems in AI adoption, as well as the importance of change management for the accomplishment of performance outcomes related to AI adoption (Smit et al., 2024). Another empirical research within the context of South Africa reveals the influence of managerial perceptions regarding AI adoption on the accomplishment of performance outcomes within the context of human resource management as a field of AI application.

Collectively, the global evidence base shows that AI coding assistants can improve productivity, and the organizational evidence base shows that sustained ROI depends on alignment, effective governance, and culture. The evidence from Africa and South Africa also shows that context plays an important role in the success of technology adoption. Therefore, the evidence base provides justification for an in-depth investigation to explore the role of organizational culture and leadership in the achievement of ROI with AI coding assistants in software development teams in South Africa.

Research Objectives

0.1 Main Research Objective

To develop an integrated framework linking financial metrics, organizational culture, and leadership to assess and enhance the return on investment of AI coding assistants in South African software engineering teams.

0.1.1 Sub Research Objectives

1. To identify and operationalize the financial costs and benefits that define return on investment for AI coding assistants.
2. To examine how organizational culture attributes influence productivity and quality outcomes from AI coding assistants.
3. To assess which leadership practices most strongly affect return on investment outcomes.
4. To apply and validate an integrated return on investment framework within large South African organizations.
5. To formulate practical recommendations that enable managers to maximize return on investment through cultural and leadership interventions.

Research Question(s)

0.2 Main Research Question

How do organizational culture and leadership shape the return on investment of AI coding assistants in South African software engineering teams?

0.2.1 Sub Research Questions

1. What financial costs and benefits define return on investment for AI coding assistants in software engineering teams?

2. How do organizational culture attributes influence productivity and quality outcomes from AI coding assistants?

3. Which of the leadership behaviours strongly affect return on investment outcomes from AI coding assistants?

4. How can financial and organizational factors be integrated into a practical return on investment framework for large South African organizations?

5. What practical guidance can be offered to managers to maximize returns from AI coding assistance investments through cultural and leadership interventions?

Keywords: AI, ROI, GenAi.

Significance/rationale

Generative AI has the potential to revolutionize the field of software engineering by increasing productivity as well as the quality of the output in tasks related to knowledge work (Brynjolfsson et al., 2025; Peng et al., 2023). Coding assistants using AI improve the speed as well as the probability of completing tasks by developers. However, the relationship between the productivity of using AI coding assistants and the ROI of software engineering teams is not well understood.

Leadership and organizational culture have been identified as factors that affect the successful adoption of the associated benefits of using AI technology. It affects innovation through guiding organizational resources towards innovative practices (Hossain et al., 2025). A culture of learning, cooperation, and innovation is essential for organizational digital capability (Abdul Wahab & Radmehr, 2024; Li et al., 2025; Poisat et al., 2024). A system perspective, adaptability, is essential for organizational assimilation of digital technologies. Emerging markets' digital performance is dependent on their level of readiness and leadership context (Gillespie et al., 2025; Nurlia et al., 2023).

However, a link is still missing as the literature on AI coding assistants does not show a link between organizational ROI and AI coding assistants' productivity. Similarly, the literature on AI value does not show a link between organizational ROI and AI value. The literature on organizational culture and leadership does not show a link between organizational ROI and AI coding assistants' value for software engineering teams. This study closes this gap as it explores the influence of organizational culture and leadership on ROI realization from AI coding assistants for software engineering teams.

Scope/Delimitation

This study is guided by a quantitative research approach that seeks to explore the impact of organizational culture and leadership on the achievement of return on investment (ROI) for AI coding assistants in software engineering teams. The study is limited to the usage of generative AI tools that facilitate the inclusion of code, inline suggestions, debugging, and documentation, as they are part of the software development process. Literature has already proven that the usage of such tools increases productivity as well as the efficiency of task completion (Brynjolfsson et al., 2025; Noy & Zhang, 2023; Peng et al., 2023).

The target population is software engineering teams working for medium-to-large-scale organizations in South Africa who have prior experience working with AI coding assistants for more than six months. This is a reasonable time frame for evaluating the outcomes since the adoption of AI coding assistants is dependent on firm-level characteristics (Babina et al., 2024). Moreover, the digital readiness of the African continent is recognized as an environmental factor (Gillespie et al., 2025).

The researcher has decided to keep the scope of the independent variables limited to organizational culture and leadership, as they might impact the achievement of return on investment for AI coding assistants. The organizational culture is analysed by taking the learning orientation, collaboration, and innovation support as factors (Abdul Wahab & Radmehr, 2024; Fosso Wamba et al., 2024). Leadership is analysed by taking the strategic direction, governance, and digital support as factors (An et al., 2024; Hossain et al., 2025; Murire, 2024). The return on investment is analysed by taking the quantified performance indicators as factors.

Delimitation

Because the technical efficiency of AI tools has already been studied in experiments (Peng et al., 2023), it is important to highlight that this research does not focus on the accuracy benchmarking, technical algorithmic evaluation of the efficiency of the AI coding assistants, architecture of the AI model, or other technical evaluations of the efficiency. While in this research, there was a choice to focus on the socio-organizational factors of value realization in the context of AI. The research only includes software engineering teams, not other professional groups using generative AI, such as marketing, finance, or legal teams. This is because the software engineering workflow is significantly different from other professions.

The importance of organizational culture and leadership was emphasized in the research as the main independent variables. Other variables that have the potential to impact ROI have not been studied in detail, such as national regulations, the macro environment, and pricing models. Though these variables have a significant impact on ROI, the aim of the research was to concentrate on the internal variables.

Brief Limitations

There are some limitations that are built into the study. Firstly, the calculation of the ROI can depend in part on self-reported perceptions of productivity and quality. Secondly, the study's focus on South African organizations limits the study's statistical generalizability to other geographic regions. Lastly, the study's cross-sectional or short-term longitudinal design may limit the capacity to generalize to long-term financial outcomes because the impact of organizational culture and leadership on performance can develop gradually over time (An et al., 2024).

Key concepts / Definitions

Table 1: Your Table Caption Go Here

Term	Definition
AI	Artificial Intelligence
ROI	Return on Investment
SEM	Structural equation modelling

Theoretical / Conceptual framework

This study is based on the following premise: while the productivity gains derived from the use of AI coding assistants can be significant for developers, the value that can be derived from the productivity gains for the organization itself remains an open question. This view is consistent with the value realization framework, which argues that business value depends on how the organization manages, supports, and integrates the technology into its day-to-day business processes. This view also has parallels with the sociotechnical systems view, which argues that the value generated using AI coding assistants depends on the individuals, processes, leaders, and organizational structures that come together.

There is a significant body of research that suggests that generative AI has a positive impact on individual performance. Research conducted in the field has shown that generative AI has a positive effect on the productivity and quality of performance (Brynjolfsson et al., 2025). Experimental research has also shown that generative AI has a positive effect on the efficiency and quality of performance (Noy & Zhang, 2023). Research has shown that programmers working with AI assistants perform their tasks more efficiently and with better quality than those programmers who do not use AI assistants (Peng et al., 2023). However, research conducted at the firm level has shown that the ROI generated by AI is firm-specific, depending upon how the firm invests in AI (Babina et al., 2024).

In this context, leadership and organizational culture are identified as key factors that explain the disparities identified. Leadership is responsible for guiding and directing the firm, as well as the way AI coding assistants are used within the firm. The empirical evidence suggests that leadership is a key factor that affects digital or AI performance outcomes (Hossain et al., 2025; Murire, 2024). On the other hand, organizational culture is responsible for influencing the behavioural aspects of individuals within the firm. Learning culture is a culture that promotes the use of AI tools within the firm through collaboration and innovation. The empirical evidence suggests that a digital organizational culture strengthens the relationship between AI performance outcomes and organizational performance (Fosso Wamba et al., 2024). Furthermore, a digital organizational culture increases innovation outcomes by human-AI capabilities (An et al., 2024).

Accordingly, the conceptual model proposes that ROI from AI coding assistants used within software engineering teams is the key outcome variable, while leadership and organizational culture are identified as the key drivers of this outcome. The way the AI tool is integrated into the organization's workflows and standardized is the conduit variable linking leadership and organizational culture to the key outcome variable of ROI from AI coding assistants. Organizational readiness is included as a control variable because of the differing digital performance outcomes across studies (Gillespie et al., 2025; Nurlia et al., 2023). Accordingly, while AI tools improve individual performance outcomes significantly, the achievement of associated financial returns depends on leadership and organizational culture.

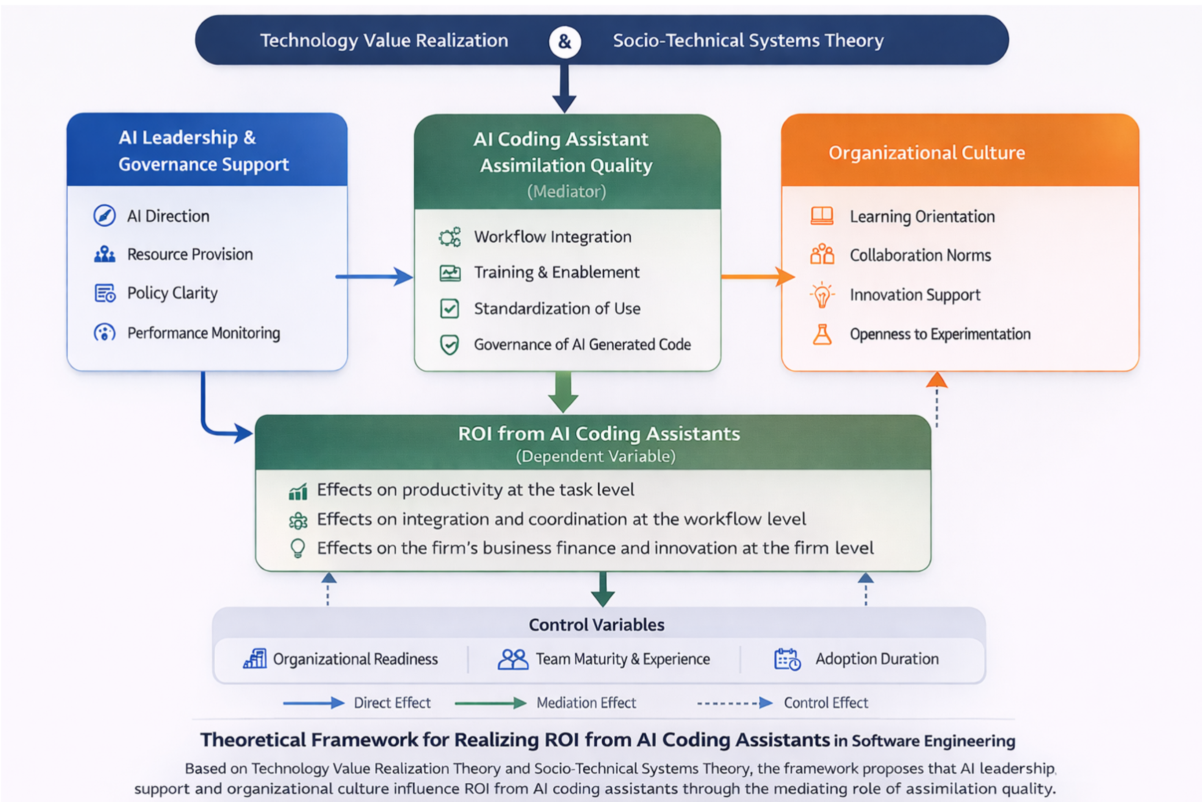

Figure 1: theoretical framework.

Declaration

I, Celumusa H Zulu, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

Celumusa H Zulu

(No signature required)

Dedication

To my Mother,

who always believe in me,

and to all my Siblings

who always believe in me

Acknowledgements

First and foremost, infinite gratitude and appreciation to my Creator, Who has given me the will to start studying again and the endurance to see this through in a time when my world was turned on its head. My perseverance, accomplishments and very being are only by Your will.

You are the very foundation my success is built on.

My deepest gratitude to my Mother, Sister and my brother's as well for always remembering me in your prayers.

My sincerest appreciation to my Supervisor Dr. Siyabonga Mhlongo, who has always been my light through this journey. You got me started on this, thank you for seeing me through to the finish line.

I'd also like to extend my sincerest appreciation to all the people that took their precious time and completed the survey to make this study possible.

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Introduction

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1.1 Section Heading

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1.1.1 Subsection Heading

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1.1.2 Some maths

LaTeX is very good at presenting mathematics.

Inline equation : $ax^2 + bx + c = 0$

Display equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Numbered equation:

$$\frac{\partial}{\partial t} U + \nabla \cdot F = 0 \tag{1.1}$$

Aligned equation (not how equals signs line up):

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \sum_{i=1}^{n} a_i b_i \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n; \end{aligned} \tag{1.2}$$

Aligned equation with no numbers:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \sum_{i=1}^{n} a_i b_i \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n; \end{aligned}$$

1.1.3 Theorems, proofs, definitions and examples

Theorem 1.1 (Fermat's last theorem)**.** *No three positive integers a, b, and c satisfy the equation $a^n + b^n = c^n$ for any integer value of n greater than 2.*

Proof. Left as an exercise for the reader. □

Definition 1.1. *The intersection of two sets A and B, denoted by $A \cap B$, is the set of all objects that are members of both the sets A and B.*

Example 1.1

Given the two sets $A = \{x : x \in \mathbb{N}, x < 5\}$ and $B = \{x : x \in \mathbb{N}, x \text{ is even}\}$ then $A \cap B = \{2, 4\}$.

1.2 Figures and tables

Figure 1.1: This is a figure caption, note how it appears underneath the figure.

Table 1.1: This is a table caption, not how it appears above the table.

first column	second column	third column
This column is left aligned	This column is centrally aligned	This column is right aligned

1.2.1 Program code

Listing 1.1: A MATLAB function to compute the first n numbers of the Fibonacci series

```matlab
function y = fibonacci(n)

% This function calculates the first n terms in the Fibonacci series

y(1) = 0;
y(2) = 1;

for i = 3 : n
    y(i) = y(i-2) + y(i-1)
end

end
```

1.3 Referencing

For additional information on citation commands, consult the `biblatex` cheat sheet via this URL: https://tug.ctan.org/info/biblatex-cheatsheet/biblatex-cheatsheet.pdf. The comprehensive `biblatex-apa` archive can be found here: https://ctan.org/pkg/biblatex-apa.

Cross referencing is easily done by using the label of the item you are referencing to (see source code for details).

- Chapter 1
- Section 1.1
- Equation (1.1)
- Table 1.1
- Figure 1.1
- Page 2

Above is an example of a bulleted list. You can also create numbered lists:

1. First list item
2. Second list item
   (a) First sub list item
   (b) Second sub list item
3. Third list item

1.4 List of Acronyms

Acronyms can be introduced using the `\gls{}` command. This works in conjunction with what is included in the `acronyms.tex` file. For example, my simple acronym (MSA) is an entry in this file, and MSA can be used throughout the document. See https://www.overleaf.com/learn/latex/Glossaries for more usage details.

Chapter 2

Literature Review

2.1 Introduction

Generative AI is a structural inflection point in knowledge-intensive work. It is different from other forms of AI and automation because the former primarily focuses on complex cognitive work through the creation of outputs that are sensitive to context and require interpretation, refinement, and integration. In the field of software engineering, AI coding assistants in the form of generative code completion tools are embedded into the coding environment and allow for the generation, debugging, documentation, and refactoring of code. Empirical studies of the impact of generative AI on human productivity and output quality revealed that, under conditions of structured task design, a significant increase in human output is achieved. In a large-scale experimental design, the availability of generative AI tools resulted in a reduction in task completion time and an increase in output quality, as determined by the researcher. The results achieved were statistically significant for the total sample, while the results were more pronounced for low-performing individuals (Noy & Zhang, 2023). The results support the idea that generative AI tools result in micro-level productivity effects. However, the productivity gains at the task level do not necessarily translate into organizational return on investment. Research on AI investment at the firm level indicates that the performance effects are heterogeneous and conditional. AI investment is linked with growth through innovation and product development channels; however, the level and stability of the returns depend on the capabilities of the organization (Babina et al., 2024). The market reaction to AI investment announcements is mixed, indicating the level of awareness among investors about the risk and capabilities (Lui et al., 2021).

These studies all circle back to the same basic idea: AI capability does not produce value; rather, value is produced when the technological tools are embedded in the organizational systems. More recent studies have suggested another component: a digital organizational culture can influence the innovation outcome of AI capability (An et al., 2024). Leadership can also play a part in influencing the development of trust and embedding AI usage (Schneider et al., 2023).

Despite the recent increase in research on the performance effects of AI, little research has appeared that combines the effects of organizational culture and leadership into a structured ROI model for AI coding assistants in software development teams. This chapter synthesizes global empirical research on AI productivity and firm performance, critically evaluates the impact of organizational conditions, and focuses the research question more specifically on the South African market, where empirical ROI research has yet to be conducted.

2.2 Generative AI and Productivity: From Task Performance to Organisational Outcomes

The best evidence for how generative AI changes work comes from tightly controlled experiments. In the case of Noy and Zhang (2023), the evidence provides strong causal proof of how gen AI increases the rate of task completion and enhances the quality of professional writing. From this, we can see precisely how the presence of gen AI changes the gap in performance. The benefits of using gen AI are not equally distributed, with the less skilled benefiting the most. This indicates the ability of gen AI to level the playing field for teams.

This uneven benefit to different individuals can, therefore, have implications for software development teams. If the benefits to gen AI for coding tasks are greater for the less skilled, the composition of the team might change. However, the benefits to the team must not only be seen at the task level. They must be sustained, integrated, and monetized to increase the ROI.

This distinction finds corroboration in firm-level evidence. Babina et al. (2024) show a positive association between the adoption of AI and firm growth and innovation, with the Association is stronger for firms with complementary capabilities. The channel through which this association seems to occur is mainly through product innovation rather than labour substitution effects, implying the importance of AI in value creation through product development rather than cost reduction.

At the same time, there is evidence of mixed reactions in the financial markets to AI investment announcements. Lui et al. (2021) show that some firms have positive abnormal returns around the announcement of AI investments, whereas others have neutral or negative returns. This appears to reflect investors' ability to distinguish between firms with the ability to successfully implement AI and those with a risk of implementation failure.

In the context of software engineering, empirical evidence points to the fact that there is a great deal of usage of AI coding assistants. However, this usage is done in an inconsistent manner. In a comprehensive empirical study done to examine the experiences of software developers, it was found that there was usage of AI coding assistants for various activities in the software development process. However, issues of trust, boundaries, and governance ambiguity have been found to hinder the effective impact of this usage (Sergeyuk et al., 2025). This points to the fact that the frequency of usage alone may not guarantee the achievement of benefits.

According to the literature, the return on investment (ROI) model of AI coding assistants comprises a three-tier structure:

2.2.1 Effects on productivity at the task level.

Task-level productivity effects relate to the observed improvements that accrue when individual programmers utilize AI-based coding assistants for the accomplishment of specific programming tasks. At the task level, AI-based coding assistants are used for activities such as code writing, documentation, debugging, and suggestion of solutions to programming problems. Such activities have the potential to hasten the accomplishment of specific tasks through the provision of real-time support during the entire process. According to experimental research, generative AI-based tools have the capability to hasten the execution of knowledge-based tasks by a large factor without compromising the quality of the output (Noy & Zhang, 2023). This implies that, at the organizational level, the ability of the development team to hasten the accomplishment of specific coding tasks has the capability to hasten the accomplishment of more complex development tasks.

2.2.2 Effects on integration and coordination at the workflow level.

Workflow-level integration and coordination effects refer to the impact of AI coding assistants in the collaboration and coordination of teams in the development of software applications. Software development involves the implementation of workflows that include code reviews, testing procedures, version control, and collaboration between different developers. The incorporation of AI coding assistants in the development process can have positive effects on the coordination and collaboration of teams in the development process. This means that the teams can be able to achieve the required objectives in the most efficient manner possible. The benefits of using AI coding assistants have been examined in the experiences of developers using the system, and the findings suggest that the benefits of the system depend on the extent of its integration in the development process (Sergeyuk et al., 2025).

2.2.3 Effects on the firm's business finance and innovation at the firm level.

Firm-level financial and innovation results refer to the overall organizational benefits that are realized when the benefits of increased productivity and workflow improvements are realized at the financial level. At the firm level, the organization seeks to determine whether the implementation of AI technologies results in benefits such as improved operational efficiency, lower development costs, faster time-to-market, or new forms of digital products/services, among others. Empirical studies conducted to determine the impact of AI technology adoption by organizations revealed that artificial intelligence technologies could drive organizational growth through innovation and improved productivity (Babina et al., 2024). Therefore, it is critical to understand the role of organizational culture and leadership in the successful implementation of AI coding assistants, given the fact that the achievement of a financial return is dependent on the organization's ability to adopt the new technology.

The majority of the literature is found to have been limited to the first tier of the ROI model. However, the second and third tiers of the model have been found to be underdeveloped in the context of AI coding assistants. This underdevelopment of the second and third tiers of the model points to the fact that the effective usage and integration of AI coding assistants have been found to be mediating factors in the ROI model.

2.3 Measuring ROI in AI Contexts: Financial, Operational, and Strategic Dimensions

Return on investment (ROI) is not absolute in nature but rather relative. ROI is generally described as net benefits earned divided by total costs incurred over a particular period. For AI initiatives, costs generally include subscription fees, integration costs, training costs, governance costs, different models used to mitigate risks, and workflow costs.

The potential benefits that could be earned from implementing AI are as follows:

2.3.1 Time saved translates to reduced labour costs.

A major advantage of AI-based coding assistants is the potential for these tools to decrease the time it takes for a programmer to do their job. If the time it takes to do development tasks is decreased, it implies that the labour costs associated with the development process will decrease. According to experimental research conducted on generative AI, the availability of AI-based assistance has a significant impact on the time it takes to complete a given knowledge-intensive task without compromising the quality of the output (Noy Zhang, 2023). Similar results have been obtained by other researchers in the field of software engineering, where it was found that the availability of AI-based assistance for the development process can accelerate the development process (Peng et al., 2023).

2.3.2 Increased productivity.

The other possible benefit associated with the use of AI coding assistants is related to the potential for enhancing development throughput. Throughput refers to the volume of work that a development team can process within a particular period. The potential for enhancing development throughput means that an organization can process a larger volume of software features, bug fixes, or updates within a single development cycle without a corresponding increase in the number of developers. Studies on the use of AI-based work tools indicate that generative AI tools have the potential to increase output levels substantially within a particular period, a factor that allows workers to process more deliverables within a particular period (Noy Zhang, 2023).

2.3.3 Defects reduced.

The coding assistants that AI offers may be beneficial in improving the quality of software codes. Defect rates refer to the number of defects or errors that are associated with a given software. It is suggested that when a developer is offered coding assistance and intelligent coding suggestions, he or she may be able to avoid certain coding defects. Research studies have shown that AI-assisted software development is beneficial to developers since it offers coding accuracy and assistance in solving coding challenges. AI-assisted software development is beneficial since it offers coding accuracy and assistance in solving coding challenges (Peng et al., 2023).

2.3.4 Reduction in rework costs.

Rework occurs when software development teams are forced to dedicate additional resources to correcting errors, re-coding, and correcting defects that are detected after the development process. It is one of the key factors that contribute to costs in software development, considering its effect on development time and its ability to cause delays in software development. By improving coding precision and offering real-time support to software developers, there is an opportunity for coding assistants to reduce the need for rework. The timely detection of errors in the development process allows software development teams to avoid the significant costs that are likely to result from addressing errors in subsequent software development phases.

2.3.5 Faster time to market.

Another way in which the use of AI coding assistants may help reduce the time it takes to deliver new software products or features is through the reduction of the time to market. Time to market refers to the length of time it takes to develop a given software product or feature from the start of the project to the release of the feature to the market. The faster the time to market, the better the organization can respond to market opportunities. The empirical literature has proven that the use of AI in the business environment leads to faster innovation processes in the business environment (Babina et al., 2024).

2.3.6 Increased revenues via innovation.

Finally, the adoption of AI has the potential to lead to organizational growth through innovation. This is because when an organization can build new features more easily or try new ideas more easily, it has the potential to create more revenue opportunities through digital products or services. The adoption of AI has the potential to lead to innovation-driven growth. The study on the adoption of AI by firms revealed that firms that adopted AI had a positive relationship between product innovation and financial performance (Babina et al., 2024). This means that an AI coding assistant has the potential to lead to not only increased operational efficiency but also increased competitiveness of an organization.

Empirical research on AI adoption in organizations suggests that AI capability is a factor in innovation-driven growth (Babina et al., 2024). For quantifying the potential benefits that could be earned from the adoption of AI in team-level software development, a more refined measurement of AI benefits is required.

In the software development domain, cycle time, deployment frequency, defect density, and mean time to recovery act as a framework to assess the overall performance in the software development domain. These could be easily translated into monetary values using the cost of delay, defect costs, and resource allocation.

The research on AI has generally used perception-based productivity metrics. Even in experiments, researchers have focused on short-term results rather than long-term monetary results (Noy Zhang, 2023).

The ideal model for calculating ROI on AI should include: ¿ Performance metrics. The objective engineering performance metrics are the measurable parameters used to evaluate the performance and productivity of software development teams. The importance of the performance metrics is undeniable since it provides empirical evidence of the performance of the AI coding assistants. The performance metrics used include the development cycle time, the deployment frequency, the defect density, and the mean time to recovery. The performance metrics are widely used to evaluate the performance of the software development process since it encompasses the productivity and quality aspects of the process. The performance of the software development process is essential since it provides evidence of the effectiveness of the AI coding assistants. For instance, the use of generative AI tools has been observed to improve the performance of the task completion speed and the quality of the outcome, which may be reflected in the performance of the software development process (Noy Zhang, 2023). The importance of the performance metrics is undeniable since it provides empirical evidence of the performance of the AI coding assistants. ¿ Cost calculations Full cost accounting is a comprehensive measure of all costs associated with the adoption and maintenance of artificial intelligence technology. In terms of AI coding assistants, organizations must consider all costs beyond the costs associated with the software tool. These costs may include costs associated with training developers, tool integration, updating policies and procedures, and maintaining the environment that is necessary to support AI usage. There are also costs associated with oversight that organizations must consider when using AI technology. Research studies have shown that organizations tend to underestimate all costs associated with technology adoption when studies are based solely on direct technology costs (Babina et al., 2024). To properly assess ROI, all costs associated with AI adoption must be considered. These costs include direct and indirect costs. ¿ Quality of organizational adoption Organisational adoption quality relates to the extent and quality of the incorporation of AI tools in organisational routines and practices. It should be noted that the mere incorporation of AI coding assistants in the development environment does not guarantee the quality and extent of the usage of such tools in organisational settings. Organisational adoption quality of AI tools can be regarded as high if developers effectively incorporate AI tools in coding activities, testing activities, and collaborative activities such as code reviews. Empirical studies of the organisational adoption of AI tools suggest that the organisational benefits of AI technologies depend significantly on the extent and quality of the incorporation of these tools in organisational routines and activities and the extent of organisational learning (An et al., 2024). Therefore, organisational adoption quality can be regarded as a critical factor in determining the organisational value of AI coding assistants. ¿ Governance costs Governance costs refer to the costs associated with managing and regulating the usage of artificial intelligence (AI) technologies in an organization. The usage of AI coding assistants poses various governance challenges to an organization in terms of data privacy, intellectual property rights, software quality assurance, and compliance with organizational policies. According to various scholarly research on the governance of AI, effective governance mechanisms are necessary to ensure that the usage of AI systems is within acceptable risk parameters and complies with organizational standards (Schneider et al., 2023). However, implementing governance mechanisms requires time and administrative costs, which are additional costs to consider when evaluating the return on investment in AI. ¿ Risks involved The risk exposure adjustments relate to how potential risks associated with the adoption of AI are considered when evaluating financial returns. There is a potential risk of incorporating coding assistants that may lead to software defects, security breaches, or copyright issues. The adoption of AI is likely to increase software defects, security breaches, or copyright issues. This implies that an organisation is likely to incur additional costs when risks associated with AI adoption are realised. According to an empirical study on organisational adoption of AI, technological advantages need to be balanced with risk management and governance to ensure sustainable value creation (Schneider et al., 2023). Adjusting the ROI formula to incorporate potential risks associated with AI adoption is essential to avoid overestimating potential benefits.

The lack of an integrated model in AI-based coding assistant research is a knowledge gap that this research aims to bridge.

2.4 Organizational Culture as a Mediator of AI Value Realisation.

Thus, organizational culture plays a significant role in the interpretation, trust, and institutionalization of AI tools. Digital organizational culture has been proven to enhance the link between AI capabilities and innovation performance (An et al., 2024). In other words, the moderating effect shows that organizational culture plays a significant role in the payoff of technological capabilities.

Trust formation is a significant factor for the adoption of AI tools. Daly et al., (2025) proved that the formation of trust with AI tools is a function of organizational experience and the presence of support systems and governance systems. Excessive trust is a threat to the adoption of AI tools, while a lack of trust is a barrier to the adoption of AI tools.

Psychological safety is a significant factor for the adoption of AI tools in the context of software engineering teams. In other words, the presence or absence of psychological safety is a significant factor for the transformation of AI suggestions into quality code.

While organizational culture is a significant factor for the adoption of AI tools, it is not quantified in the context of the AI ROI model. Most studies on the impact of organizational culture on AI adoption qualitatively address the impact of organizational culture on AI adoption. Thus, the present study proposes a quantified approach to organizational culture as a mediating factor for the impact of AI coding assistants on the ROI.

2.5 Leadership and AI Governance.

The leadership function assumes critical importance in ensuring that AI adoption is in sync with business goals and risk management processes. The body of research on AI governance has identified three interrelated but distinct layers of AI governance data governance, model governance, and organizational governance (Schneider et al., 2023). However, leadership ensures that AI governance structures are embedded within the organization rather than being adopted in an informal and ad hoc manner.

The leadership function assumes importance in defining return on investment (ROI) in AI in the following areas:

2.5.1 Executive sponsorship.

Leadership has a pivotal role to play in ensuring the achievement of value from artificial intelligence technology. Research studies on the digital transformation have all proven the importance of leadership in the achievement of the objectives of technology adoption. Among the most important leadership constructs in the achievement of technology adoption objectives is executive sponsorship, which refers to the support role played by the organizational leadership in the achievement of the objectives of technology adoption. When the organizational leadership provides support to the objectives of technology adoption, it becomes easier to integrate emerging technology, such as the use of artificial intelligence coding assistants, into the organizational strategy instead of it being used as a standalone productivity tool by individual developers (Schneider et al., 2023). Further research studies on artificial intelligence technology have proven the importance of the organizational leadership in the achievement of organizational preparedness to adopt digital innovations and the ability of employees to integrate artificial intelligence technology into their work practices (An et al., 2024).

2.5.2 Resource allocation.

Another important role of a leader involves resource allocation, which includes the provisioning of the financial, technical, and human resources needed to support the effective integration of the various AI systems into the organizational processes. The effective adoption of AI systems in the organizational processes is more than just the acquisition of the various AI systems. For instance, empirical research on the economic impact of the adoption of AI systems has shown that the ability of the organization to invest in the acquisition of various capabilities would have a positive impact on the ability of the organization to realize productivity improvements from the various AI systems (Babina et al., 2024). In the case of software engineering, the allocation of resources may limit the ability of the developer to experiment with the various coding assistants, thus limiting the ability to realize the various learning processes.

2.5.3 Metrics alignment.

Leadership has a vital role to play in the achievement of metrics alignment, which refers to the alignment of organizational performance metrics and the anticipated outcomes of implementing AI systems. The coding assistants of AI systems influence various aspects of software development performance, including development speed, code quality, collaboration, and defect reduction. Studies on the productivity outcomes of generative AI systems suggest that, while generative AI systems are instrumental in expediting task completion, the performance outcomes are contingent on the evaluation and integration of the output with the organizational process (Noy Zhang, 2023). In cases where speed is the main objective and not adequately balanced with quality outcomes, the adoption of generative AI systems may inadvertently result in an escalation of technical debt and defects. However, metrics that balance speed with quality outcomes allow an organization to leverage the performance outcomes of generative AI systems to achieve operational performance.

2.5.4 Policy enforcement.

Another key responsibility of leaders in this case is policy enforcement, which is described as the development and implementation of governance systems that guide the usage of AI systems in the workflow of an organisation. The coding assistants in AI systems pose new challenges in terms of governance, including intellectual property rights and verification of the source of the code, and managing risks to security. The research on AI governance shows that when an organisation has a system of governance in place, it is in a better position to manage technological risks and sustain performance improvements with the adoption of AI systems (Schneider et al., 2023). This reduces uncertainty among developers and provides guidance on the usage of AI systems.

2.5.5 Ethical considerations.

Finally, the ethical considerations associated with the use of artificial intelligence need to be addressed by the leaders. In the case of generative artificial intelligence, ethical considerations arise regarding transparency, ownership of intellectual properties, data privacy, and bias. Ethical governance practices ensure the effective implementation of artificial intelligence technology in a manner that is consistent with the regulatory framework and ethical codes of practice. Ethical artificial intelligence practices help to build trust relationships with employees, which is a critical factor in the long-term institutionalization of artificial intelligence technology. Empirical research studies on the organizational responses to artificial intelligence technology have established the importance of ethical governance mechanisms in building trust relationships between employees and artificial intelligence technology, which is essential to sustain the integration of artificial intelligence technology into the organization (Daly et al., 2025).

If leadership were to be absent in AI adoption, then AI coding assistants would be used in an optional capacity rather than being used to drive business performance. In addition to this, it would also help in reducing risks in intellectual property and code authenticity.

However, research studies that have investigated leadership in AI adoption and its direct linkage with financial ROI in AI coding assistants are scarce. The body of research has primarily been cantered on AI governance principles but has seldom been able to quantify financial ROI.

This has led to the need to incorporate leadership as a key mediating variable in AI ROI modelling.

2.6 South African Context and Emerging Market Considerations.

The process of AI adoption in EMs takes place in an institutional environment where capability heterogeneity and resource scarcity exist. In South Africa, for instance, the organizational stakeholders of the country's software engineering industry acknowledge the competitive benefits and labour adjustments of AI adoption (Poisat et al., 2024).

South Africa's software engineering industry comprises large organizations with intricate systems and a strong demand for digital transformation. However, the governance level and the allocation of workforce competencies differ among organizations. This highlights the importance of leadership and organizational culture in the achievement of AI benefits. In addition, in resource-scarce environments, justifying the return on investment (ROI) of AI implementation becomes essential. In this case, misalignment in the organization increases the risk of implementation.

Despite the surge in interest in the adoption of AI in South African software engineering teams, there still exists a scarcity of peer-reviewed literature on the ROI of using AI coding assistants. This highlights the importance of this paper.

2.7 Synthesis and Research Gap.

However, current studies indicate that generative AI models boost task-level productivity (Noy Zhang, 2023). Moreover, AI investments are related to firm expansion and innovation outcomes (Babina et al., 2024). On the contrary, market reactions to AI adoption exhibit heterogeneity, reflecting risks associated with AI adoption, like AI integration (Lui et al., 2022). Furthermore, AI model performance is conditional upon organizational culture in the digital space (An et al., 2024). The role of AI governance is emphasized, requiring alignment between leadership and strategic goals (Schneider et al., 2023). In addition, contextual limitations exist in the South African environment (Poisat et al., 2024).

Nevertheless, some critical knowledge gaps remain: ¿ Insufficient return on investment modelling for AI-based coding assistants The current literature on the subject suggests that AI coding assistants can improve the productivity of developers. However, there is a lack of literature on the financial return on investment of AI coding assistants. Most of the literature focuses on the productivity of developers at the task level or the perceived usefulness of AI tools. For example, experimental studies have found that generative AI tools significantly improve the efficiency of task performance and the quality of outputs in a knowledge work setting (Noy Zhang, 2023). Other studies on AI programming tools have found that developers can perform coding tasks quickly with the aid of AI tools (Peng et al., 2023). However, there is a lack of literature on the financial return on investment of AI coding assistants. ¿ Inadequate inclusion of culture and leadership within the financial model Organisational culture and leadership have been recognised as key factors in the achievement of successful technology implementation outcomes. The empirical findings have suggested that the organisational culture for digitalisation can strengthen the impact of the capabilities of artificial intelligence on innovation and performance (An et al., 2024). Leadership has been observed as the key factor in the development of the governance structure for the implementation of responsible AI in the organisational environment (Schneider et al., 2023). However, the importance of organisational culture and leadership has been recognised in the literature while studying these factors in isolation for the achievement of technological performance outcomes. Only limited empirical research has been conducted on the development of the organisational culture and leadership factors within the financial framework for the assessment of the return on investment in AI technologies. ¿ Insufficient measurement of the operation at the team level The available literature on AI-driven productivity tends to focus more on individual performance measures and not team-level measures to ensure operational effectiveness. However, it is to be noted that software development is a team-based activity in which developers interact with each other through various processes. Studies that have been conducted to understand the role of AI coding assistants have shown that these tools are used at various stages of the software development lifecycle, though there is a significant variation in their usage (Sergeyuk et al., 2024). It is not clear how these tools influence the productivity and quality of software engineering teams without analysing team-level measures. ¿ Lack of empirical research from the continent Most of the research on AI adoption and its organizational outcomes is based on empirical research undertaken in North America, Europe, and parts of Asia. Therefore, there is a scarcity of empirical research on how these AI technologies impact organizational outcomes in Africa. Research on the impact of artificial intelligence on the workforce in South Africa has seen a surge of interest in artificial intelligence adoption while underscoring the need to conduct more research on its impact on organizational outcomes within the local environment (Poisat et al., 2024). The scarcity of research on AI coding assistants on software engineering teams within South Africa impedes our understanding of how contextual factors impact the realization of the ROI on AI.

This research intends to fill the knowledge gap by developing and empirically testing an integrated model that connects culture, leadership, effective adoption, operation, and measurable return on investment for software development teams in South Africa.

Chapter 3

Research Methodology

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Chapter 4

Results

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Chapter 5

Discussion

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Chapter 6

Conclusion

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Appendix A

Ethical Clearance Certificate

UNIVERSITY OF JOHANNESBURG

COLLEGE OF BUSINESS AND ECONOMICS RESEARCH ETHICS COMMITTEE (CBEREC)

Dear CH Zulu,

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that the proposed research project indicated in the table below, has been reviewed by the School of Consumer Intelligence and Information Systems (SCiiSREC) Research Ethics Committee of the University of Johannesburg (UJ). Ethical clearance is hereby granted and is valid for three years, from 1 May 2026 until 30 April 2029.

Applicant	CH Zulu
Supervisor/s	Dr S Mhlongo
Student/staff number	226166283
Title	The role of IT leadership in realising the return on investment of AI coding assistants
Decision date at meeting	30 April 2026
Reviewers	SCIISREC 2/9 RV1 V2 AISREC Dr F Radebe; Dr S Dube
Ethical clearance code	2026SCiiS024
Rating of application	CODE 01
CODE 01 – Approved	CODE 02 – Approved with suggestions/requirements with no re-submission
CODE 03 – Referred back	CODE 04 – Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. Researcher/s, i.e. internal and/or external applicants, who wish to undertake research at the University of Johannesburg (UJ) case institution as data collection source involving UJ students and/or UJ staff and/or any UJ support services submit the Personal Information Impact Assessment (PIIA) via the 2026 UJ Gatekeeper Platform, after familiarising themselves with the UJ POPIA and PAIA manual and, if applicable, utilising the UJ login to register POPI requests.

2. The researcher/s understand the implications of any form of data breach and takes responsibility for any possible violation or any form of unethical research conduct. The onus is on the researcher/s to ensure that the project adheres to ethical research requirements.

3. The researcher/s will be conducting the study as set out in the approval application, i.e. Form_7a and/or Form_7b linked to the departmentally approved research proposal that was submitted by the researcher/s, together with supporting documents, primary data collection, and/or secondary data analysis, to the research ethics committee.
4. The researcher/s will ensure that the project adheres to all applicable legislation, scopes of practice, professional codes of conduct, and scientific standards as it pertains to the field or study.
5. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and/or unexpected ethical management issues.
6. Any changes that can affect the study-related risks for participants or researchers must be reported immediately to the research ethics committee in writing.
7. No data collection or fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the research ethics committee.
8. The researcher/s must be aware that infectious diseases may influence researcher/s, research participants, and responses. Steps must be taken to the best of researcher/s' ability to avoid the spread of infectious diseases.
9. Researcher/s familiarise themselves with the PAIA and POPIA UJ compliance manual. All data collection must be done within Government regulations, for instance, researcher/s must uphold all provisions of the Protection of Personal Information Act (POPIA) Act 4 of 2013 when collecting, processing, and storing data. This includes but is not limited to upholding the principles of accountability, processing restriction, purpose, transparency, processing restrictions, security, and rights of participants.
10. Researcher/s abide by the ethical use of Artificial Intelligence (AI). As non-legal entities, AI tools cannot assert the presence or absence of conflicts of interest nor manage copyright and license agreements. UJ requires researcher/s to disclose any use of AI in the research process and the preparation of their research report to ensure transparency and allow for proper assessment by reviewers and examiners.

The School of Consumer Intelligence and Information Systems Research Ethics Committee wish the researcher/s all the best with this research project.

Kind regards,

TdPlessis

Prof T du Plessis
SCiiSREC Chair
tduplessis@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

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Appendix C

Another Appendix Chapter

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